



Medical Policy

Subject: Cardiac Contractility Modulation Therapy

Document #: SURG.00153

Status: Reviewed

Publish Date: 10/01/2025

Last Review Date:
08/07/2025

Description/Scope

This document addresses the use of cardiac contractility modulation therapy designed to treat chronic moderate-to-severe heart failure.

Position Statement

Investigational and Not Medically Necessary:

The use of cardiac contractility modulation therapy is considered **investigational and not medically necessary** for all indications, including but not limited to heart failure.

Rationale

Summary

Cardiac contractility modulation (CCM) therapy has been proposed as a treatment for individuals with moderate-to-severe heart failure (HF). In March 2019, the U.S. Food and Drug Administration (FDA) granted premarket approval (P180036) to Impulse Dynamics for the OPTIMIZER[®] Smart Implantable Pulse Generator (Impulse Dynamics, Orangeburg, NY), following a breakthrough device designation in 2015. It was approved for the treatment of individuals with chronic, moderate-to-severe HF (New York Heart Association [NYHA] Class III or ambulatory Class IV) who remain symptomatic despite guideline-directed medical therapy (GDMT). FDA required that recipients be in normal sinus rhythm with left ventricular ejection fraction (LVEF) from 25% to 45% and not considered candidates for cardiac resynchronization therapy (CRT) to restore normal heart rhythm. The OPTIMIZER Smart System treatment, referred to as CCM, delivers electrical signals to the ventricles during the ventricular absolute refractory period. The expected result is improvement in the 6-minute walk test distance, quality of life, functional status, and exercise tolerance (FDA Product Information Label, 2019). Current randomized evidence for CCM therapy is limited in size, predominantly unblinded, and has not demonstrated consistent improvements in key outcomes such as mortality or heart-failure hospitalization.

Discussion

Clinical Trial Development and Regulatory Approval

The original technology for CCM was developed by Impulse Dynamics. The original device was evaluated in a trial by the FDA that did not demonstrate efficacy. The original FIX-HF-5 study used a broader group of HF participants and endpoints that were difficult to achieve in the clinical trial. Ultimately, the trial failed, but subgroup analysis of the FIX-HF-5 study showed which participants could possibly benefit from use of the device.

The FIX-HF-5 study was a phase II, prospective, unblinded, randomized study comparing CCM plus optimal medical treatment (OMT) to OMT alone in 428 participants. Participants had NYHA functional class III or IV HF with LVEF of $\leq 45\%$ (Abraham, 2015). The FIX-HF-5 study met its primary safety endpoint, but did not reach its primary efficacy endpoint, changes in ventilatory anaerobic threshold of responders. However, significant improvements in primary and secondary efficacy endpoints, including the responders' ventilatory anaerobic threshold endpoint, were met in a prespecified FIX-HF-5 subgroup analysis of individuals with LVEF ranging from 25%-45% ($n=221$; $p=0.001$). Based on the subgroup analysis, a new, prospective study was designed and conducted to confirm the efficacy of CCM in this population.

Abraham and colleagues (2018) reported results from the FIX-HF-5C confirmatory study to prospectively test the efficacy and safety of CCM in participants with NYHA functional class III or IV symptoms and LVEF 25-45%. A total of 160 participants were randomized to continue OMT ($n=86$) or CCM (treatment, $n=74$, unblinded) for 24 weeks. The primary efficacy endpoint was met with a peak oxygen uptake between groups of 0.84 (95% Bayesian credible interval: 0.123 to 1.552) ml O₂/kg/min. The following secondary outcomes, Minnesota Living With Heart Failure questionnaire (MLWHFQ) ($p<0.001$), NYHA functional class ($p<0.001$), and 6-minute walk test ($p=0.02$), were all significantly better in the treatment

group compared to the control group. Of the 68 individuals (n=68/74) in the treatment group who underwent implantation with the OPTIMIZER device, seven device-related events were reported. There was a reduction in the composite of cardiovascular death and HF hospitalizations from 10.8% to 2.9% (p=0.048). There was one death related to sepsis at 164 days after device implantation in the treatment group following surgery for an incarcerated hernia. Long-term risks of infection cannot be known at this time. Additional studies are needed to confirm the benefit outside of this small study population and beyond the 24-week follow-up in this study.

The FDA approval of the OPTIMIZER Smart System was based on findings reported for 389 participants with moderate-to-severe HF (FIX-HF-5 subgroup, n=229; FIX-HF-5C, n=160) in two randomized, multicenter clinical trials published in peer-reviewed journals (Abraham, 2018; Kadish, 2011). Participants received OMT alone or OMT plus implantation with an OPTIMIZER Smart System. The study design included criteria such as NYHA function class III or ambulatory class IV with HF despite OMT, an LVEF ranging from 25-45% as determined by echocardiographic core laboratory, and normal sinus rhythm with QRS duration < 130 ms. Individuals who had an LVEF ≤ 35% were required to have an implantable cardioverter defibrillator (ICD) unless there were contraindications.

The primary effectiveness endpoint was met, with an estimated mean difference in peak oxygen uptake at 24 weeks between the CCM groups and control groups of 0.84 mL/kg/min with a Bayesian credible interval of (0.12, 1.55) mL/kg/min. FDA stated that "the probability that CCM is superior to control was 0.989, which exceeds the 0.975 criterion required for statistical significance of the primary endpoint." Among the pooled data, 60.1% (104/173) of participants in the CCM group and 34.9% (59/169) in the control group achieved ≥ 1 class improvement in NYHA at 24-week follow up period. At the 24-week follow-up, the change in Quality of Life measured by the MLWHFQ total score between the groups in the pooled data, was -10.9 (95% confidence interval; -14.6, -7.2). The primary safety endpoint was met. "The complication-free proportion in the CCM group cohort was 89.7% (61/68) with lower confidence limit of 79.9% (one-sided alpha=0.025), which was greater than the pre-defined threshold of 70%. The majority of complications (5/7, 71.4%) were lead dislodgements." Among the OPTIMIZER group and the control group, the freedom from death (98.3%, 95.3%; p=0.2549), cardiovascular death (100%, 96.5%; p=0.1198), composite rate of all-cause death or all-cause hospitalizations (78.1%, 77.7%; p=0.9437) and overall rate of adverse events and serious adverse events were similar at 24-weeks (FDA Product Information Label, 2019).

FDA Regulatory Assessment

According to the FDA Summary of Safety and Effectiveness Data, the agency concluded that even though the primary effectiveness endpoint (change in peak oxygen uptake) met its pre-specified endpoint, the clinical significance was questioned, primarily because the observed treatment difference was due to a decline from baseline in the control arm. The treatment arm, depending on the analysis method, either showed a decline in peak oxygen uptake or a marginal increase, making claims of increased exercise tolerance not justifiable. Two subjective endpoints, Quality of life per the MLWHFQ and the 6-minute walk test, did show an improvement. However, the confidence intervals were wide, possibly due to the relatively small sample size and unblinded nature of the trial (the control group did not receive a device). The latter raised the possibility that the positive outcomes for the subjective endpoints could be due to a placebo effect.

Real-World Clinical Experience

Observational studies have provided additional insights into the real-world effectiveness of CCM therapy. A single-center study Deak (2025) of 31 consecutive participants with NYHA class III HF showed significant mid-term improvements after CCM implantation, with 68% of participants experiencing improvement in NYHA functional class (mean improvement of 0.97 classes, p<0.001). The study also demonstrated a reduction in mean annualized hospitalizations from 0.8 to 0.4 per year (p=0.048) and an 8% improvement in LVEF (from 32% to 40%, p=0.002) over a mean follow-up of 1.4 years. Additionally, participants experienced significant weight loss, averaging 8.5 pounds (4% of body weight), with 77% of participants losing weight. However, the study also noted procedural complications in 16% of participants, primarily related to pain, with complications more common in early implants.

A multicenter registry study by Davtyan (2024) of 166 participants with heart failure with reduced ejection fraction found that CCM therapy efficacy may vary by HF etiology. Participants with non-ischemic cardiomyopathy showed significantly greater improvements in LVEF (+7.7% vs. +2.0%, p<0.001) and more pronounced reverse remodeling compared to those with ischemic cardiomyopathy. Despite these functional differences, 12-month mortality was similar between groups (10.8% vs. 14.4%, p=0.51), suggesting that CCM therapy is equally safe across both etiologies.

In a prospective registry of 503 participants (Kuschyk, 2021), continuous CCM delivered by the OPTIMIZER Smart system was associated with reported 24-month benefits in NYHA class (mean -0.6 ± 0.7), MLWHFQ score (-10 ± 21), and LVEF (+5.6 ± 8.4 percentage-points; all p<0.001). It should be noted that the confidence intervals for these observed results all cross the zero value, indicating the true change could be improvement, no improvement, or worsening. Heart-failure hospitalizations fell from 0.74 to 0.25 events per patient-year (p<0.0001). The study was unblinded and used historical controls, so the hospitalization rate may have been influenced by knowledge of the participants' treatments as well as time-based confounding. The observed one- and three-year survival exceeded mortality predicted by the Meta-Analysis Global Group in Chronic Heart Failure risk score, except in participants with LVEF ≤ 25%. Although 100% source verification was performed, the absence of a parallel control arm means that pre-/post comparisons remain vulnerable to secular trends and survivor bias. Other potentially confounding factors include the fact that echocardiographic follow-up occurred only when clinically indicated, and rhythm (atrial fibrillation vs normal sinus rhythm) or medication changes were not randomized or

tracked. Investigator ties to the manufacturer also warrant caution. Additional information from controlled studies are needed to demonstrate a cause and effect benefit from CCM treatment.

Long-term retrospective safety data regarding renal function have emerged from a 5-year observational study of 187 CCM recipients treated at a single center in Germany (Yuecel, 2024). The study found that kidney function and chronic kidney disease stage distribution remained generally stable over 60 months of follow-up, with only minor changes in estimated glomerular filtration rate (from 58.2 to 54.2 mL/min/1.73 m², $p < 0.05$) occurring at the 5-year mark. Participants experienced improvements in cardiac function regardless of their baseline kidney function, though those with advanced chronic kidney disease had higher rates of HF hospitalizations. Renal function was monitored using eGFR (CKD-EPI), creatinine, BUN, uric acid, and electrolytes. Urinary markers such as albumin and proteinuria were not assessed, which limits proper CKD staging for many participants. The absence of a comparison group of heart failure patients who did not receive CCM prevents the study from establishing whether CCM directly caused the observed outcomes.

Future multicenter, prospective trials with comprehensive renal phenotyping are needed to confirm this study's findings.

A comparative study between CCM ($n=105$) and CRT with defibrillator ($n=220$) found that both therapies produced similar 12-month improvements in functional status and LVEF. However, CCM participants had significantly higher HF hospitalization rates (45.7 vs. 16.8 events per participant-year, odds ratio 4.2, $p < 0.001$), which may be attributed to more advanced HF at baseline in the CCM group (lower LVEF: 23.6% vs. 26.3%, worse NYHA class: 3.03 vs. 2.94, both $p < 0.05$). The study concluded that both devices can provide comparable functional improvements when appropriately selected (Yuecel, 2025).

Professional Society Guidelines and Meta-Analyses

These individual trials and real-world studies have collectively informed subsequent meta-analyses and professional society positions on CCM therapy. Giallauria and colleagues (2020) reported findings from a comprehensive meta-analysis of individual data from all known randomized trials. The authors reviewed the effects of CCM therapy on functional capacity and HF-related quality of life and found CCM promising. The authors concluded that "larger, well-conducted randomized controlled trials using a parallel double-blind design are needed in order to determine the effect of CCM on major mortality and morbidity outcomes before CCM can be widely recommended as an effective treatment option for HF patients."

In 2021, the European Society of Cardiology (ESC) guidelines for the diagnosis and treatment of acute and chronic HF listed CCM as a device under evaluation for individuals with NYHA class III-IV HF (McDonagh, 2021). The committee concluded that the device was associated with small improvement in exercise tolerance and quality of life.

The 2022 American Heart Association (AHA)/American College of Cardiology (ACC)/Heart Failure Society of America (HFSA) Guideline for the management of heart failure does not include recommendations for the use of CCM therapy as a treatment for HF (Heidenreich, 2022).

The 2025 ACC/AHA Appropriate Use Criteria (AUC) for implantable cardioverter-defibrillators, cardiac resynchronization therapy, and pacing provides guidance based on expert consensus, a process distinct from the formally graded recommendations found in major clinical practice guidelines. This represents the first time CCM has been formally addressed in appropriate use criteria. Based on consensus derived through a modified Delphi process, the guideline authors rated CCM as "May Be Appropriate" across various NYHA classes (II to IV) and LVEF ranges (<25%, 25%-35%, 36%-45%) for individuals not eligible for CRT (QRS duration <130 ms). These ratings reflect potential benefits in functional status and quality of life but underscore significant evidence gaps, as the AUC emphasizes that CCM lacks strong randomized data on hard outcomes like mortality or hospitalization reduction. The authors stated that:

In addition, there are currently no practice guideline recommendations for CCM. The lack of differentiation of AUC recommendations according to HF class as well as the paucity of U.S. or European guideline recommendations likely reflect the strength of evidence related to hard clinical outcomes, such as HF hospitalization and mortality. Studies evaluating longer-term clinical outcomes and impact of CCM on reverse remodeling in larger cohorts are needed.

Ongoing Research and Future Directions

The OPTIMIZER[®] Smart System is being studied in an ongoing prospective, multicenter, post approval study (NCT03970343). The study is designed to evaluate the long-term safety and efficacy of the device as well as to rule out placebo effects and more precisely identify the group of individuals that most benefit from the device. Enrollment criteria are limited to individuals with NYHA functional class III symptoms and a left ventricular EF of 25-45%. The study-estimated enrollment is 620 participants, with estimated study completion date in March 2026.

The OPTIMIZER Integra CCM-D System (CCM-D System) is an implantable cardiac device system that combines an implantable cardioverter defibrillator (ICD) and CCM into one device. It is being studied in an ongoing, single-arm, prospective, multicenter study with 300 participants who will be followed for at least 2 years and has an estimated study completion date of December 2025 (NCT05855135). The study aims to assess how safely and effectively the CCM-D can convert:

Induced ventricular fibrillation (VF) and spontaneous ventricular tachycardia and/or ventricular fibrillation (VT/VF) episodes in subjects with Stage C or D heart failure who remain symptomatic despite being on guideline-directed medical therapy (GDMT), are not indicated for cardiac resynchronization therapy (CRT) and have heart failure with reduced left ventricular ejection fraction (LVEF $\leq 40\%$).

Since the OPTIMIZER Integra CCM-D System incorporates a CCM device as an integral component, it is subject to the same medical necessity considerations outlined above for other CCM devices.

Background/Overview

According to the Centers for Disease Control and Prevention (CDC) nearly 6.7 million Americans are currently diagnosed with HF (CDC, 2024; based on 2022 statistics from the American Heart Association). Approximately 50% of individuals with HF die within 5 years of diagnosis. As a result of HF, the weakened heart muscle causes inadequate filling of the left ventricle, as well as a backflow of blood into the left atrium, both resulting in decreased cardiac output and increased symptoms for the afflicted individual. Symptoms can include shortness of breath, fatigue, swelling in the ankles, feet, legs, abdomen, and veins in the neck. Currently there is no cure for HF; medical therapy includes a combination of diuretics, digoxin, angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARB), beta-blockers, and aldosterone antagonists. Some individuals may remain symptomatic, despite medical therapy. Ongoing studies evaluate other treatment options to assist physicians in the management of individuals with severe HF.

Definitions

Bayesian hierarchical analysis: A statistical method providing estimates of post-analysis parameters based on frequencies observed in a prior analysis evaluated in a series of hierarchical models.

Guideline-directed medical therapy (GDMT): This term was adopted by the writing groups for the major specialty medical societies, (such as found in Tracy, 2012 and Yancy, 2013) in 2012; the term replaces and is synonymous with "Optimal medical therapy."

Heart failure (HF): A condition in which the heart no longer adequately functions as a pump. As blood flow out of the heart slows, blood returning to the heart through the veins backs up, causing congestion in the lungs and other organs.

New York Heart Association (NYHA) Definitions:

The NYHA classification of heart failure is a 4-tier system that categorizes participants based on subjective impression of the degree of functional compromise; the four NYHA functional classes are as follows:

- **Class I** - individuals with cardiac disease but without resulting limitation of physical activity; ordinary physical activity does not cause undue fatigue, palpitation, dyspnea, or anginal pain; symptoms only occur on severe exertion.
- **Class II** - individuals with cardiac disease resulting in slight limitation of physical activity; they are comfortable at rest; ordinary physical activity (for example, moderate physical exertion such as carrying shopping bags up several flights of stairs) results in fatigue, palpitation, dyspnea, or anginal pain.
- **Class III** - individuals with cardiac disease resulting in marked limitation of physical activity; they are comfortable at rest; less than ordinary activity causes fatigue, palpitation, dyspnea or anginal pain.
- **Class IV** - individuals with cardiac disease resulting in inability to carry on any physical activity without discomfort; symptoms of heart failure or the anginal syndrome may be present even at rest; if any physical activity is undertaken, discomfort is increased.

Ventilatory Anaerobic Threshold (VAT): The point in exercise testing at which anaerobic metabolism is detected by comparing oxygen consumption with CO₂ production. VAT provides a measure of exercise capacity that has prognostic value for individuals with heart failure.

VO₂: Oxygen uptake. This can be calculated using the Fick Equation ($VO_2 = [SV \times HR] \times [CaO_2 - CvO_2]$) in which oxygen uptake equals stroke volume times the heart rate times the difference in oxygen concentration between arterial and venous blood. VO₂ indicates functional aerobic capacity is widely used as a measure of cardiorespiratory fitness.

Coding

The following codes for treatments and procedures applicable to this document are included below for informational purposes. Inclusion or exclusion of a procedure, diagnosis or device code(s) does not constitute or imply member coverage or provider reimbursement policy. Please refer to the member's contract benefits in effect at the time of service to determine coverage or non-coverage of these services as it applies to an individual member.

When services are Investigational and Not Medically Necessary:

For the following procedure codes or when the code describes a procedure indicated in the Position Statement section as investigational and not medically necessary.

CPT

0408T	Insertion or replacement of permanent cardiac contractility modulation system, including contractility evaluation when performed, and programming of sensing and therapeutic parameters; pulse generator with transvenous electrodes
0409T	Insertion or replacement of permanent cardiac contractility modulation system, including contractility evaluation when performed, and programming of sensing and therapeutic parameters; pulse generator only
0410T	Insertion or replacement of permanent cardiac contractility modulation system, including contractility evaluation when performed, and programming of sensing and therapeutic parameters; atrial electrode only
0411T	Insertion or replacement of permanent cardiac contractility modulation system, including contractility evaluation when performed, and programming of sensing and therapeutic parameters; ventricular electrode only
0412T	Removal of permanent cardiac contractility modulation system; pulse generator only
0413T	Removal of permanent cardiac contractility modulation system; transvenous electrode (atrial or ventricular)
0414T	Removal and replacement of permanent cardiac contractility modulation system pulse generator only
0415T	Repositioning of previously implanted cardiac contractility modulation transvenous electrode, (atrial or ventricular lead)
0416T	Relocation of skin pocket for implanted cardiac contractility modulation pulse generator
0417T	Programming device evaluation (in person) with iterative adjustment of the implantable device to test the function of the device and select optimal permanent programmed values with analysis, including review and report, implantable cardiac contractility modulation system
0418T	Interrogation device evaluation (in person) with analysis, review and report, includes connection, recording and disconnection per patient encounter; implantable cardiac contractility modulation system
	Note the following CPT codes will be effective for services on or after 01/01/2025:
0915T	Insertion of permanent cardiac contractility modulation-defibrillation system component(s), including fluoroscopic guidance, and evaluation and programming of sensing and therapeutic parameters; pulse generator and dual transvenous electrodes/leads (pacing and defibrillation)
0916T	Insertion of permanent cardiac contractility modulation-defibrillation system component(s), including fluoroscopic guidance, and evaluation and programming of sensing and therapeutic parameters; pulse generator only
0917T	Insertion of permanent cardiac contractility modulation-defibrillation system component(s), including fluoroscopic guidance, and evaluation and programming of sensing and therapeutic parameters; single transvenous lead (pacing or defibrillation) only
0918T	Insertion of permanent cardiac contractility modulation-defibrillation system component(s), including fluoroscopic guidance, and evaluation and programming of sensing and therapeutic parameters; dual transvenous leads (pacing and defibrillation) only
0919T	Removal of a permanent cardiac contractility modulation-defibrillation system component(s); pulse generator only
0920T	Removal of a permanent cardiac contractility modulation-defibrillation system component(s); single transvenous pacing lead only
0921T	Removal of a permanent cardiac contractility modulation-defibrillation system component(s); single transvenous defibrillation lead only
0922T	Removal of a permanent cardiac contractility modulation-defibrillation system component(s); dual (pacing and defibrillation) transvenous leads only
0923T	Removal and replacement of permanent cardiac contractility modulation-defibrillation pulse generator only
0924T	Repositioning of previously implanted cardiac contractility modulation-defibrillation transvenous electrode(s)/lead(s), including fluoroscopic guidance and programming of sensing and therapeutic parameters
0925T	Relocation of skin pocket for implanted cardiac contractility modulation-defibrillation pulse generator
0926T	Programming device evaluation (in person) with iterative adjustment of the implantable device to test the function of the device and select optimal permanent programmed values with analysis, including review and report, implantable cardiac contractility modulation-defibrillation system
0927T	Interrogation device evaluation (in person) with analysis, review, and report, including connection, recording, and disconnection, per patient encounter, implantable cardiac contractility modulation-defibrillation system
0928T	Interrogation device evaluation (remote), up to 90 days, cardiac contractility modulation-defibrillation system with interim analysis and report(s) by a physician or other qualified health care professional

0929T	Interrogation device evaluation (remote), up to 90 days, cardiac contractility modulation-defibrillation system, remote data acquisition(s), receipt of transmissions, technician review, technical support, and distribution of results
0930T	Electrophysiologic evaluation of cardiac contractility modulation-defibrillator leads, including defibrillation-threshold evaluation (induction of arrhythmia, evaluation of sensing and therapy for arrhythmia termination), at time of initial implantation or replacement with testing of cardiac contractility modulation-defibrillator pulse generator
0931T	Electrophysiologic evaluation of cardiac contractility modulation-defibrillator leads, including defibrillation-threshold evaluation (induction of arrhythmia, evaluation of sensing and therapy for arrhythmia termination), separate from initial implantation or replacement with testing of cardiac contractility modulation-defibrillator pulse generator
0948T	Interrogation device evaluation (remote), up to 90 days, cardiac contractility modulation system with interim analysis, review and report(s) by a physician or other qualified health care professional
0949T	Interrogation device evaluation (remote), up to 90 days, cardiac contractility modulation system, remote data acquisition(s), receipt of transmissions, technician review, technical support, and distribution of results

HCPCS

C1824	Generator, cardiac contractility modulation (implantable)
K1030	External recharging system for battery (internal) for use with implanted cardiac contractility modulation generator, replacement only

ICD-10 Procedure

02H63MZ	Insertion of cardiac lead into right atrium, percutaneous approach [when specified as a lead for a contractility modulation device]
02HK3MZ	Insertion of cardiac lead into right ventricle, percutaneous approach [when specified as a lead for a contractility modulation device]
0JH60AZ	Insertion of contractility modulation device into chest subcutaneous tissue and fascia, open approach
0JH63AZ	Insertion of contractility modulation device into chest subcutaneous tissue and fascia, percutaneous approach
0JH80AZ	Insertion of contractility modulation device into abdomen subcutaneous tissue and fascia, open approach
0JH83AZ	Insertion of contractility modulation device into abdomen subcutaneous tissue and fascia, percutaneous approach

ICD-10 Diagnosis

All diagnoses

References

Peer Reviewed Publications:

1. Abraham WT, Kuck KH, Goldsmith RL, et al. A randomized controlled trial to evaluate the safety and efficacy of cardiac contractility modulation. *JACC Heart Fail.* 2018; 6(10):874-883.
2. Abraham WT, Lindenfeld J, Reddy VY, et al. A randomized controlled trial to evaluate the safety and efficacy of cardiac contractility modulation in patients with moderately reduced left ventricular ejection fraction and a narrow QRS duration: study rationale and design. *J Card Fail.* 2015; 21(1):16-23.
3. Anker SD, Borggrefe M, Neuser H, et al. Cardiac contractility modulation improves long-term survival and hospitalizations in heart failure with reduced ejection fraction. *Eur J Heart Fail.* 2019; 21(9):1103-1113.
4. Davtyan K, Chugunov I, Topchyan A, et al. HF Etiology and cardiac contractility modulation therapy. *BMC Cardiovasc Disord.* 2024; 24(1):279.
5. Deak A, Zaidi SM, Gangireddy C, et al. Mid-term clinical outcomes and cardiac function in patients receiving cardiac contractility modulation. *J Interv Card Electrophysiol.* 2025; 68(3):579-588.
6. Giallauria F, Cuomo G, Parlato A, et al. A comprehensive individual patient data meta-analysis of the effects of cardiac contractility modulation on functional capacity and heart failure-related quality of life. *ESC Heart Fail.* 2020; 7(5):2922-2932.
7. Kadish A, Nademanee K, Volosin K, et al. A randomized controlled trial evaluating the safety and efficacy of cardiac contractility modulation in advanced heart failure. *Am Heart J.* 2011; 161(2):329-337.
8. Kloppe A, Mijic D, Schiedat F, et al. A randomized comparison of 5 versus 12 hours per day of cardiac contractility modulation treatment for heart failure patients: a preliminary report. *Cardiol J.* 2016; 23(1):114-119.
9. Kuschyk J, Falk P, Demming T, et al. Long-term clinical experience with cardiac contractility modulation therapy delivered by the Optimizer Smart system. *Eur J Heart Fail.* 2021; 23(7):1160-1169.
10. Müller D, Rempis A, Schauerte P, et al. Clinical effects of long-term cardiac contractility modulation (CCM) in subjects with heart failure caused by left ventricular systolic dysfunction. *Clin Res Cardiol.* 2017; 106(11):893-904.

11. Ponikowski P, Francis DP, Piepoli MF, et al. Enhanced ventilator response to exercise in patients with chronic heart failure and preserved exercise tolerance: maker of abnormal cardiorespiratory reflex control and predictor of poor prognosis. *Circulation*. 2001; 103(7):967-972.
12. Yucel G, Gaasch L, Kodeih A, et al. Device-therapy in chronic heart failure: cardiac contractility modulation versus cardiac resynchronization therapy. *ESC Heart Fail*. 2025;12(1):456-466.
13. Yucel G, Yazdani B, Schreiner K, et al. Long-Term Renal Function with cardiac contractility modulation therapy. *Cardiorenal Med*. 2024; 14(1):385-396.
14. Wiegand P, Chan R, Jost C, et al. Safety, performance, and efficacy of cardiac contractility modulation delivered by the 2-lead OPTIMIZER Smart System: the FIX-HF-5C2 Study. *Circ Heart Fail*. 2020; 13(4):e006512.

Government Agency, Medical Society, and Other Authoritative Publications:

1. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA Guideline for the management of heart failure. A report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Amer Coll Cardiol*. 2022; 79:e263-e421.
2. Impulse Dynamics. Assessment of Combined CCM and ICD Device in HFrEF (INTEGRA-D). NLM Identifier: NCT05855135. Last updated June 10, 2025. Available at: <https://www.clinicaltrials.gov/study/NCT05855135?cond=ccm-d&rank=1>. Accessed on July 13, 2025.
3. Impulse Dynamics. Post Approval Study (PAS) of the OPTIMIZER Smart and CCM therapy (PAS). NLM Identifier: NCT03970343. Last updated May 7, 2025. Available at: <https://www.clinicaltrials.gov/ct2/show/NCT03970343?term=CCM+Therapy+AND+OPTIMIZER&cond=Heart+Failure&draw=2&rank=1>. Accessed on June 10, 2025.
4. McDonagh TA, Metra M, Adamo M, et al. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. Developed by the Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC). *Eur. Heart J*. 2021; 75(6):3599-3726.
5. Mozaffarian D, Benjamin EJ, Go AS, et al. AHA Statistical Update. Heart disease and stroke statistics - 2016 update a report from the American Heart Association. *Circulation*. 2015; 133:e38-e360.
6. Russo AM, Desai MY, Do MM, et al. ACC/AHA/ASE/HFSA/HRS/SCAI/SCCT/SCMR 2025 appropriate use criteria for implantable cardioverter-defibrillators, cardiac resynchronization therapy, and pacing: a report of the American College of Cardiology Solution Set Oversight Committee, American Heart Association, American Society of Echocardiography, Heart Failure Society of America, Heart Rhythm Society, Society for Cardiovascular Angiography and Interventions, Society of Cardiovascular Computed Tomography, and Society for Cardiovascular Magnetic Resonance. *J Am Coll Cardiol*. 2025; 85(11):1213-1285.
7. Tracy CM, Epstein AE, Darbar D, et al. 2012 ACCF/AHA/HRS focused update of the 2008 guidelines for device-based therapy of cardiac rhythm abnormalities: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *J Am Coll Cardiol*. 2012; 60(14):1297-1313.
8. U.S. Food and Drug Administration (FDA). Pre Market Approval (PMA). Medical Devices. OPTIMIZER Smart System - No. P180036. Rockville, MD: FDA. March 21, 2019. Available at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpma/pma.cfm?id=P180036>. Accessed on July 13, 2025.
9. U.S. Food and Drug Administration (FDA) Premarket Notification Database. Implantable Pulse Generator, OPTIMIZER Smart System. Summary of Safety and Effectiveness Data (SSED). No. P180036. Rockville, MD: FDA. March 21, 2019. Available at: https://www.accessdata.fda.gov/cdrh_docs/pdf18/P180036b.pdf. Accessed on July 10, 2025.
10. Yancy CW, Jessup M, Bozkurt B, et al. 2017 ACC/AHA/HFSA focused update of the 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Failure Society of America. *J Am Coll Cardiol*. 2017; 70:776-803.
11. Yancy CW, Jessup M, Bozkurt B, et al. 2013ACCF/AHA guideline for the management of heart failure. A report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2013; 128:e240-e327.

Websites for Additional Information

1. Centers for Disease Control and Prevention. Heart failure. Last updated May 15, 2024. Available at: https://www.cdc.gov/heart-disease/about/heart-failure.html?CDC_AAref_Val=https://www.cdc.gov/heartdisease/heart_failure.htm. Accessed on July 13, 2025.
2. National Heart, Lung and Blood Institute. Heart failure. Last updated March 24, 2022. Available at: http://www.nhlbi.nih.gov/health/dci/Diseases/Hf/HF_WhatIs.html. Accessed on July 10, 2025.

Index

Cardiac Contractility Modulation (CCM) Therapy
Heart Failure
OPTIMIZER Smart Implantable Pulse Generator
OPTIMIZER Smart System
OPTIMIZER Integra CCM-D System

The use of specific product names is illustrative only. It is not intended to be a recommendation of one product over another, and is not intended to represent a complete listing of all products available.

Document History

Status	Date	Action
Reviewed	08/07/2025	Medical Policy & Technology Assessment Committee (MPTAC) review. Revised Rationale, Background/Overview, Definitions, References, and Websites sections.
	07/01/2025	Updated Coding section with 07/01/2025 CPT changes, added 0948T, 0949T.
Reviewed	08/08/2024	MPTAC review. Revised Rationale, Background/Overview, Definitions, References and Websites sections. Updated Coding section with 01/01/2025 CPT changes, added 0915T-0931T effective 01/01/2025.
Reviewed	08/10/2023	MPTAC review. Updated References and Websites sections.
Reviewed	08/11/2022	MPTAC review. Updated Rationale, References and Websites sections.
	04/01/2022	Updated Coding section with 04/01/2022 HCPCS changes; added K1030.
Reviewed	08/12/2021	MPTAC review. Updated Rationale, Background, References and Websites sections.
Reviewed	08/13/2020	MPTAC review. Updated Rationale, Background, References and Websites sections.
	12/31/2019	Updated Coding section with 01/01/2020 HCPCS changes; added C1824.
Reviewed	08/22/2019	MPTAC review. Updated References and Websites sections.
New	06/06/2019	MPTAC review. Initial document development.

Federal and State law, as well as contract language, including definitions and specific contract provisions/exclusions, take precedence over Medical Policy and must be considered first in determining eligibility for coverage. The member's contract benefits in effect on the date that services are rendered must be used. Medical Policy, which addresses medical efficacy, should be considered before utilizing medical opinion in adjudication. Medical technology is constantly evolving, and we reserve the right to review and update Medical Policy periodically.

No part of this publication may be reproduced, stored in a retrieval system or transmitted, in any form or by any means, electronic, mechanical, photocopying, or otherwise, without permission from the health plan.

© CPT Only – American Medical Association

Services provided by Anthem HealthChoice HMO, Inc. and/or Anthem HealthChoice Assurance, Inc., licensees of the Blue Cross and Blue Shield Association, an association of independent Blue Cross and Blue Shield Plans.